
ME 632: Mechanics for Energy Systems

Course Code	ME 632
Course Name	Mechanics for Energy Systems
L-T-P-C	3-0-0-3
Prerequisites	Mechanics of Solid
Intended for	B. Tech./ M. Tech. /MS
Distribution	Compulsory for M. Tech in Mechanical Engineering with specialization in Energy Systems, and Elective for others
Approval	10th Senate (Jun 2016)

Course Contents

- **Brief overview of linear elasticity**

Material constitutive relations, generalized Hooke's law, problems of linear elasticity, principle of Virtual work; minimum Potential energy; Hamilton's equation [4 Lectures]

- **Stress Analysis**

Analysis of various components used in energy systems/ subsystems under different types of stresses - axial, bending and torsion; piping systems, turbine blade, rotors, boilers, compressor, nozzle, blower, generator, solar concentrating collectors and associated drives. [22 Lectures]

- **Dynamic Analysis**

Free and forced vibrations; analysis of rotor systems, geared systems; natural frequencies and natural modes, steady state response

Different kinds of bearing used in various types of turbine-rotor; stiffness and damping coefficients of journal bearings, half frequency whirl and resonance whip [10 Lectures]

- **Balancing**

Balancing of rigid and flexible turbine rotors, influence coefficient and modal balancing techniques for flexible rotors. [4 Lectures]

Text and Reference Books

1. Boresi, A.P. and Sidebottom, O.M., **Advanced Mechanics of Materials**, John Wiley, 1993.
2. Timoshenko, S.P. and Goodier, J.B., **Theory of Elasticity**, McGraw-Hill Kogakusha Ltd., 1970
3. J. S. Rao, **Rotor Dynamics**, New Age International Publishers, New Delhi.
4. Timoshenko, D H. Young and W. Weaver, **Vibration Problems in Engineering**, John Wiley.